2.1 Continued

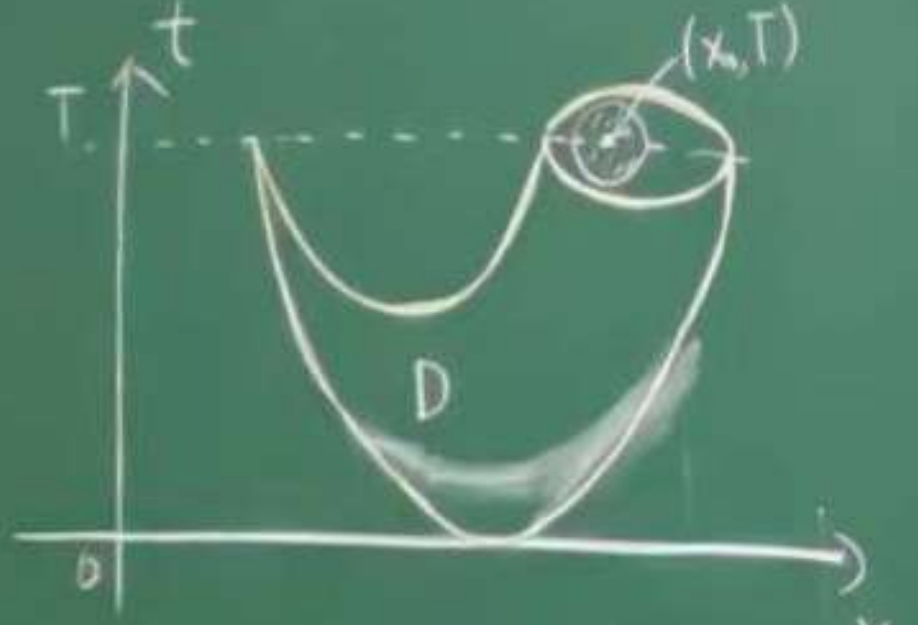
$D \subseteq \mathbb{R}^{n+1} = \{(x,t) \mid x \in \mathbb{R}^n, t \in \mathbb{R}\}$ bdd domain

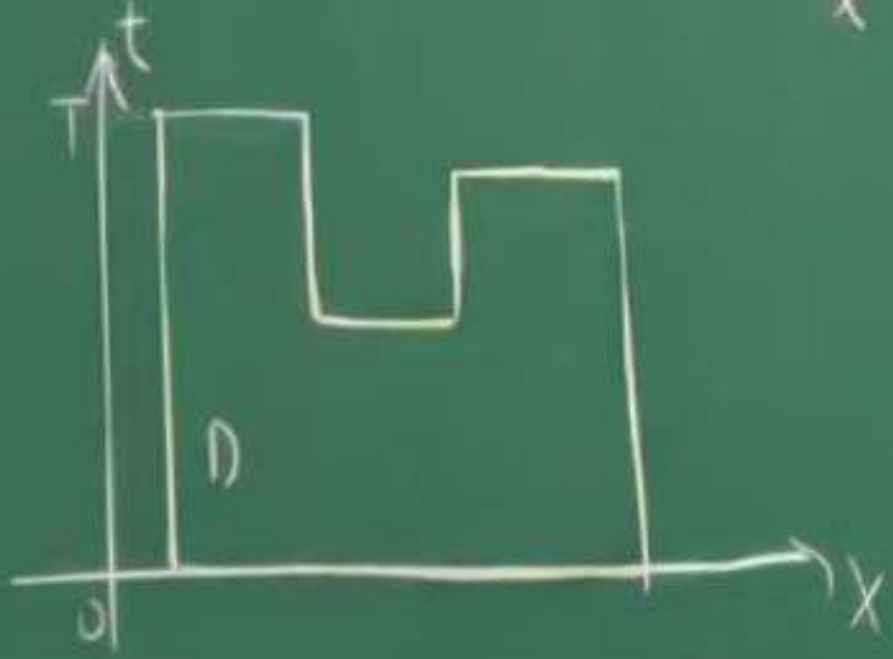
$(x,t) \in D \Rightarrow 0 < t < T$

$\bar{D} \cap \{t=0\} \neq \phi$, $\bar{D} \cap \{t=T\} \neq \phi$

Parabolic boundary: $\Gamma = \overline{\partial D \cap \{0 \leq t < T\}}$

Parabolic interior: $\bar{D} \setminus \Gamma$





Fact. $(x_0,T) \in$ Parabolic interior $D \setminus \Gamma$

$\Rightarrow \exists\, \delta > 0$ s.t.

$$B_\delta((x_0,T)) \cap D) = \underbrace{B_\delta((x_0,T)) \cap \{t < T\}}_{:= B_\delta^-}$$

pf: $\because (x_0,T) \notin \Gamma$, Γ is cpt

$\therefore \exists\, \delta > 0$ small s.t. $B_\delta((x_0,T)) \cap \Gamma = \phi$

$$B_\delta^- = \underbrace{(B_\delta^- \cap D)}_{\text{open}} \cup \underbrace{(B_\delta^- \cap D^c)}_{\text{open}}$$

$$= B_\delta^- \cap \left(\bar{D}^c \cup \Gamma \cup (\partial D \cap \{t = T\}) \right)$$

$$= \underbrace{(B_\delta^- \cap \bar{D}^c)}_{\text{open}} \cup \underbrace{(B_\delta^- \cap \Gamma)}_{\phi} \cup \underbrace{\left(B_\delta^- \cap (\partial D \cap \{t = T\}) \right)}_{\phi}$$

$(B_\delta^- \cap D) \cap (B_\delta^- \cap D^c) = \phi$

both open, B_δ^- connected

$\Longrightarrow B_\delta^- \cap D^c = \phi$, i.e., $B_\delta^- \subset D$

$(x_0, T) \in \partial D \Rightarrow B_\delta(x_0, T) \cap D \neq \phi$

$\Rightarrow B_\delta(x_0, T) \cap D = B_\delta^-$.

$\Rightarrow (x, t) \in B_\delta(x_0, T) \cap D$

$\Rightarrow \forall (x, t) \in B_\delta^- \Rightarrow B_\delta^- \cap D \neq \phi$

$t < T$,

□

Consider operator: $Lu = \frac{\partial u}{\partial t} - \left[a_{ij}(x,t) u_{x_i x_j} + b_i(x,t) u_{x_i} + c(x,t) u \right]$

Def. We say L is <u>strictly parabolic</u> in D if $\exists$ const. $\lambda_0 > 0$ s.t. $\underbrace{(a_{ij}(x,t))_{n\times n} \geq \lambda_0 I_{n\times n}}_{(x,t)\in \bar{D}\setminus\Gamma}$ $\forall\ (x,t)\in \bar{D}\setminus\Gamma$.

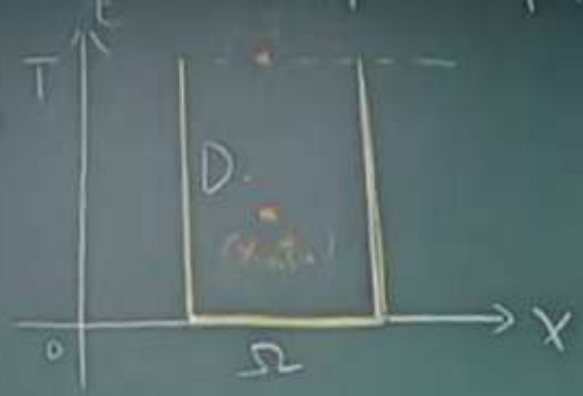
(i.e., $\sum_{i,j=1}^{n} a_{ij}(x,t)\xi_i\xi_j \geq \lambda_0 |\xi|^2$ $\forall\ (x,t)\in \bar{D}\setminus\Gamma$, $\forall\ \xi\in\mathbb{R}^n$.

L is called <u>parabolic</u> in D if $(a_{ij}(x,t))_{n\times n} > 0$ $\forall\ (x,t)\in \bar{D}\setminus\Gamma$.

<u>degenerately parabolic</u> in D if $(a_{ij}(x,t))_{n\times n} \geq 0$ $\forall\ (x,t)\in \bar{D}\setminus\Gamma$.

2.2 Weak Maximum Principle

Intuition:



$D = \Omega \times (0,T)$, $\Gamma = (\partial\Omega \times [0,T]) \cup (\Omega \times \{t=0\})$

$u_t - \Delta u = f(x,t) \leq 0$ "fridge"

expect: $\max_{\bar{D}} u = \max_{\Gamma} u$

WMP ($c \equiv 0$): Suppose L is <u>degenerately parabolic</u> in a bdd domain $D \subset \mathbb{R}^{n+1}$

If $u \in C^{2,1}(D \setminus \Gamma) \cap C^0(\bar{D})$ satisfies <u>$Lu \leq 0$</u> in $D \setminus \Gamma$,

("fridge")

then $\max_{\bar{D}} u = \max_{\Gamma} u$.

Pf: First consider special case $\underline{Lu<0}$ in $\bar{D}\setminus\Gamma$

Let $(x_0,t_0)\in\bar{D}$ s.t. $u(x_0,t_0)=\max_{\bar{D}} u$ ($\because \bar{D}$ cpt, $u\in C^0(\bar{D})$)

Case 1. $(x_0,t_0)\in\Gamma$. done.

Case 2. $(x_0,t_0)\in\bar{D}\setminus\Gamma$: $\frac{\partial u}{\partial t}(x_0,t_0)\geq 0$, $\nabla_x u(x_0,t_0)=0$;



$\left(u_{x_ix_j}(x_0,t_0)\right)_{n\times n}\leq 0$.

$$Lu(x_0,t_0)=\underbrace{\frac{\partial u}{\partial t}(x_0,t_0)}_{\geq 0}-\Big[\underbrace{a_{ij}(x_0,t_0)u_{x_ix_j}(x_0,t_0)}_{\leq 0}+\underbrace{b_i(x_0,t_0)u_{x_i}(x_0,t_0)}_{=0}\Big]\geq 0$$

$\not\Rightarrow$ $Lu<0$ in $\bar{D}\setminus\Gamma \Longrightarrow$ Case 2 cannot happen.

General case $Lu \leq 0$ in $\bar{D} \setminus \Gamma$

Let $V_\varepsilon = u - \varepsilon t$, $\varepsilon > 0$ small

$$L(V_\varepsilon) = Lu - \varepsilon L(t) = \underbrace{Lu}_{\leq 0} - \varepsilon < 0 \text{ in } \bar{D} \setminus \Gamma$$

Special case $\Rightarrow$ $\max_{\bar{D}} V_\varepsilon = \max_{\Gamma} V_\varepsilon$

Let $\varepsilon \to 0$: $\downarrow$ $\downarrow$

$$\max_{\bar{D}} u = \max_{\Gamma} u$$

□

Rk: In the pf, we only need $(a_{ij}(x,t))_{n\times n} \geq 0$ in $D\setminus\Gamma$.

In particular, when $a_{ij}\equiv 0$, $Lu = \frac{\partial u}{\partial t} - b_i(x,t)u_{x_i}$, transport equation

we still have W.M.P

WMP($c\leq 0$): Let L be degerately parabolic in $D\subseteq \mathbb{R}^{n+1}$ bdd domain.

If $u\in C^{2,1}(\bar{D}\setminus\Gamma)\cap C^0(\bar{D})$ satisfies $Lu\leq 0$ in $D\setminus\Gamma$,

then $\max_{\bar{D}} u \leq \max_{\Gamma} u^+$

Pf Case 1 $\max_{\bar{D}} u \le 0$ trivial

Case 2 $\max_{\bar{D}} u > 0$ proof is the same as WMP ($c = 0$) □

(In this case, $\max_{\bar{D}} u = \max_{\Gamma} u$)

Theorem Let L be degenerately parabolic in $D \subseteq \mathbb{R}^{n+1}$ bdd domain, with $c(x,t) \le M$. If $u \in C^{2,1}(D \cup \Gamma) \cap C^0(\bar{D})$ satisfies $Lu \le 0$ in $D \cup \Gamma$, $u|_\Gamma \le 0$, then $u \le 0$ in $\bar{D}$.

Pf: Let $V(x,t)=e^{-Mt}u(x,t)$. W.T.S. $V\le 0$ in $\bar{D}$.

$$V_t = -M\cdot e^{-Mt}\cdot u + e^{-Mt}\cdot u_t = e^{-Mt}(u_t - Mu)$$

$$V_{x_i} = e^{-Mt}u_{x_i},\quad V_{x_ix_j} = e^{-Mt}\cdot u_{x_ix_j}$$

$$V_t - [a_{ij}V_{x_ix_j} + b_iV_{x_i} + \underbrace{(c-M)}_{\le 0}V] = e^{-Mt}[u_t - Mu - a_{ij}u_{x_ix_j} - b_iu_{x_i} - (c+M)u]$$

$$= e^{-Mt}\cdot Lu \le 0 \text{ in } D\setminus\Gamma$$

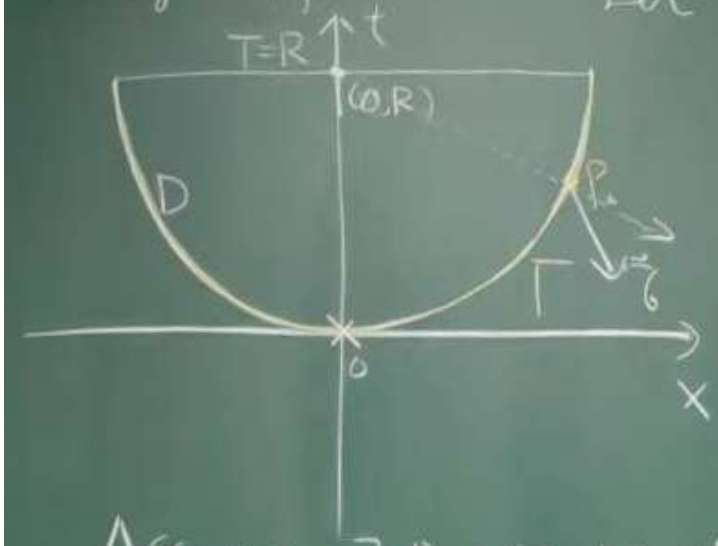
$$\xrightarrow{\text{WMP}((\le 0)} \max_{\bar{D}} V \le \max_{\Gamma} V^+ = 0 \quad (\because u\le 0 \text{ on } \Gamma \Rightarrow V\le 0 \text{ on } \Gamma)$$

$$\Rightarrow V\le 0 \text{ in } \bar{D} \Rightarrow u\le 0 \text{ in } \bar{D}$$

□

23 Hopf boundary point Lemmas

Baby Hopf B.P.L: Let $B_R^- = \{(x,t) \in \mathbb{R}^n \times \mathbb{R} \mid |x|^2 + (t-R)^2 < R^2,\ t \le 1\}$



(Parabolic interior of D)

Suppose L is strictly parabolic on B_R^- and

$\|a_{ij}, b_i, c\|_{L^\infty(D)} \le M$ Let $u \in C^{2,1}(B_R^-) \cap C^0(\overline{B_R^-})$

satisfy $Lu \le 0$ in B_R^- ("fridge")

Assume $\exists P_0 = (x_0, t_0) \in \Gamma$, $x_0 \neq 0$, $u(x_0,t_0) > u(x,t)$ $\forall (x,t) \in \overline{B_R^-} \setminus \{P_0\}$.

Then $\forall$ outward point vector $\vec{\eta}$ at P_0, we have $\frac{\partial u}{\partial \vec{\eta}}(P_0) > 0$ (if it exists)

provided (i) $c \equiv 0$, or (ii) $c \le 0$ in B_R^- and $u(P_0) \ge 0$, or (iii) $u(P_0) = 0$